The Effect of Temporary Alcohol Bans on Domestic Violence, Road Safety and Health

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Abstract

Alcohol consumption generates substantial social externalities; however, causal evidence on the broader consequences of alcohol availability remains limited, especially for domestic violence. We exploit repeated temporary alcohol bans implemented during election weekends in Ecuador to estimate the effects of restricting alcohol availability using a difference-in-differences design and ten electoral events between 2018 and 2025. Our analysis combines nationwide administrative datasets covering 15.5 million emergency calls, 172,656 traffic accidents, and 7.7 million hospital admissions. We find that temporary alcohol bans substantially reduce alcohol-related emergency calls and domestic violence, as well as traffic accidents, and external injury hospital admissions. Event-study estimates show that these effects emerge immediately after the ban takes effect, consistent with reduced alcohol availability as the underlying mechanism. The results indicate that temporary restrictions on alcohol availability generate broad social benefits by reducing externality costs, and provide causal evidence on the impact of alcohol on domestic violence, road safety, and health.

Keywords: Alcohol consumption; Domestic violence; Road crashes; Morbidity, Difference-in-differences; Public policy.

JEL Classification: I18, J10.

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1 Introduction

Alcohol consumption generates substantial social costs through its effects on physical and mental health (Rehm et al. 2009, Shield et al. 2020). Given that many of these harms are borne by individuals other than drinkers, alcohol consumption gives rise to important negative externalities and has long motivated public intervention. Among these externalities, gender-based violence is particularly important, with harmful alcohol consumption recognized as one of the leading risk factors for violence against women, including domestic and intimate partner violence (WHO 2026). Globally, approximately one in four ever-partnered women (around 682 million women) have experienced physical or sexual violence by an intimate partner during their lifetime, while, depending on the country and methodology, the economic costs of violence against women range from 0.3 to 6 percent of GDP (Commonwealth Secretariat 2022, WHO 2025*b*). Despite these substantial social costs, causal evidence on whether reducing alcohol availability lowers domestic violence remains surprisingly scarce.

Although a growing literature has examined the effects of alcohol policies on mortality, crime, and health outcomes (Carpenter & Dobkin 2009, Cook & Durrance 2013, Hansen 2015, Nelson & McNall 2016, Barron et al. 2024, Grönqvist & Niknami 2014), just a few studies have examined their effects on domestic violence (Luca et al. 2015, Ivandić et al. 2024). Existing evidence typically focuses on a single outcome or relies on a single administrative data source. In consequence, much less is known about the broader social consequences of alcohol availability, including whether temporary restrictions simultaneously reduce domestic violence, traffic accidents, and injuries. Evidence from developing and middle-income countries is particularly scarce despite the disproportionately high burden of alcohol-related harm in these settings.

This paper exploits temporary alcohol bans implemented during election weekends in Ecuador to estimate the causal effects of restricting alcohol availability on domestic violence and other alcohol-related externalities. Ecuadorian law prohibits the sale and consumption of alcoholic beverages from Friday through Sunday during national and local elections, while voting takes place only on Sunday. Using a difference-in-differences design, we compare changes in outcomes before and after the alcohol ban within election weeks with the corresponding changes during the same calendar weeks in years not affected by an election. Our empirical design exploits ten election weekends between 2018 and 2025, combining multiple treatment events with multiple control periods across Ecuador's municipalities.

We find that temporary alcohol bans generate substantial reductions across a broad range of alcohol-related harms. Alcohol-related emergency calls decline by 33 percent, providing direct evidence that the policy reduces alcohol-related behaviors. Domestic violence emergency calls fall by 28.5 percent, equivalent to approximately 73 fewer incidents nationwide each day during alcohol-ban weekends. We also find significant reductions in fatal and non-fatal traffic accidents and external injury hospital admissions. Event-study estimates show that these effects emerge immediately after the ban takes effect on Friday and persist throughout the weekend, consistent with reduced alcohol availability as the mechanism linking the policy to lower violence and injuries. Back-of-the-envelope calculations further indicate that these reductions translate into substantial short-run economic savings through avoided domestic violence and road traffic crashes.

Our analysis combines three nationwide administrative datasets. First, we use 15.5 million emergency calls recorded by Ecuador's ECU-911 emergency response system between 2021 and 2025, including more than 1.3 million alcohol-related incidents and over 430,000 domestic violence incidents. Second, we use administrative records on 172,656 recorded traffic accidents collected by the National Transit Agency between 2018 and 2025. Third, we use approximately 7.7 million hospital admissions compiled by the National Institute of Statistics and Census. Whereas previous studies typically examine a single consequence of alcohol availability, we estimate its effects across multiple administrative systems, including emergency calls, traffic accidents, and hospital admissions, which allows us to characterize the broader social externalities of alcohol consumption.

Our study contributes to the limited literature linking alcohol consumption and violence against women. Existing evidence from the United Kingdom shows that increased alcohol consumption during sporting events affects domestic violence and crimes against women (Ivandić et al. 2024).

However, the existence of a causal link between alcohol availability and domestic violence and the mechanism through which it runs has not been comprehensively studied, except for Luca et al. (2015). Exploiting the exogenous timing of temporary alcohol bans and high-frequency emergency service call data, we provide causal evidence that reduced alcohol availability lowers domestic violence. The concurrent decline in alcohol-related emergency calls supports alcohol consumption as a mechanism linking temporary alcohol bans to domestic violence. Finally, the scale and granularity of our data enable us to examine heterogeneity across socioeconomic conditions.

We also contribute to the literature on alcohol regulation and its social consequences. Previous studies have shown that changes in alcohol availability affect crime, violence, mortality, and road safety (Carpenter & Dobkin 2009, Cook & Durrance 2013, Hansen 2015, Grönqvist & Niknami 2014). Related work also documents causal effects of heavy episodic drinking on traffic accidents and hospital admissions (Francesconi & James 2019, Marcus & Siedler 2015). Closest to our setting, Nakaguma & Restrepo (2018) exploit electoral dry laws in Brazil and show that temporary alcohol bans reduce traffic accidents and hospitalization costs, focusing on election day only, while Barron et al. (2024) examine South Africa's temporary alcohol prohibition during the COVID-19 pandemic. We contribute to this literature by showing that temporary alcohol restrictions simultaneously reduce domestic violence, traffic accidents, and injury-related hospital admissions.

Methodologically, our empirical design is related to recent difference-in-differences studies exploiting high-frequency administrative data to evaluate policy changes (Leslie & Wilson 2020, Abrams 2021, Barron et al. 2024). Unlike these studies, however, our identification relies on repeated policy shocks rather than a single intervention and compares each treated election week with multiple matched control weeks from different years. This repeated-event design reduces reliance on a single episode and allows treatment effects to be identified across multiple independent policy implementations. Moreover, because the electoral alcohol bans were implemented independently of broader policy packages, our setting is less likely to be confounded by concurrent nationwide interventions, such as the multiple restrictions introduced during the COVID-19 pandemic that those other papers used.

Although our analysis focuses on Ecuador, the setting is informative for a broader class of middle-income countries characterized by high levels of violence against women and substantial alcohol-related harms (Commonwealth Secretariat 2022, WHO 2024). Ecuador combines strong enforcement of temporary alcohol bans with rich nationwide administrative records covering emergency services, recorded traffic accidents, and hospital admissions (Nazif-Muñoz 2025, Lopez Rodriguez et al. 2025). These features provide an appropriate setting for studying the broader social consequences of alcohol availability and complement recent evidence from Brazil, South Africa, India, Australia, the United Kingdom, and the United States. More broadly, our findings contribute with evidence to inform global efforts such as the World Health Organization's SAFER initiative, which identifies restrictions on alcohol availability as a cost-effective strategy for reducing alcohol-related harm (WHO 2025*a*), and complement the WHO's RESPECT framework for preventing violence against women (WHO 2026).

The remainder of the paper is organized as follows. Section II describes the institutional setting and the data. Section III presents the identification strategy and empirical framework. Section IV reports the main results on alcohol-related incidents, domestic violence, traffic accidents, and injury-related hospital admissions. Section V presents robustness checks and additional analyses. Section VI concludes.

2 Policy Context and Data

2.1 Policy Context

Ecuadorian law prohibits the sale and consumption of alcoholic beverages during national and local elections. Commonly referred to as the *Ley Seca* (dry law), the restriction begins at noon on Friday and remains in force until noon on Monday, covering the entire election weekend, while voting takes place only on Sunday (Asamblea Nacional del Ecuador 2009). Individuals violating the prohibition are subject to monetary fines and may face arrest, and businesses selling alcoholic beverages during the ban are also sanctioned. The restriction is nationwide and strongly enforced, with supermarkets, liquor stores, bars, and other establishments prohibited from selling alcoholic beverages. This makes election weekends generate sharp but temporary reductions in alcohol availability.

Between 2018 and 2025, Ecuador held ten national and local electoral events, including presidential, legislative, municipal, and referendum elections. The repeated implementation of the alcohol ban provides a sequence of shocks to alcohol availability that we exploit to estimate its short-run effects on domestic violence, road safety, and health. Table 1 summarizes the electoral events and the corresponding alcohol-ban periods. Given that electoral events rarely occur in the same calendar week across years, each treated week can be matched to the corresponding calendar week in multiple non-election years, substantially increasing the number of control observations.

Table 1. Electoral Events and Timing of Alcohol Bans

Year	Election event	Week of the year	Start of the ban	Election day
2018	Referendum	5	Feb. 2	Feb. 4
2019	Local elections	12	Mar. 22	Mar. 24
2021	General election (1st round)	6	Feb. 5	Feb. 7
2021	General election (2nd round)	15	Apr. 9	Apr. 11
2023	Referendum	6	Feb. 3	Feb. 5
2023	General election (1st round)	34	Aug. 18	Aug. 20
2023	General election (2nd round)	42	Oct. 13	Oct. 15
2024	Referendum	16	Apr. 19	Apr. 21
2025	General election (1st round)	6	Feb. 7	Feb. 9
2025	General election (2nd round)	15	Apr. 11	Apr. 13

Notes: The alcohol ban begins on Friday and remains in force through Sunday (election day). For each treated week, the control group consists of observations from the same calendar week in all years in which no elections took place during that particular week.

Although Ecuador is the empirical setting, the policy examined here belongs to a broader class of temporary restrictions on alcohol availability used in many countries, including election-related dry laws, late-night sales restrictions, and event-specific alcohol prohibitions. Understanding the effects of these policies is particularly relevant because alcohol-related harms remain substantial. According to the World Health Organization, Ecuador consumed 2.4 liters of pure alcohol per capita among individuals aged 15 years and older in 2022, a level comparable to that of many middle-income countries (WHO 2024).

Alcohol-related harms also represent major public policy challenges. According to the 2019 National Survey on Violence against Women, 64.9 percent of women aged 15 years and older have experienced at least one form of violence during their lifetime, 42.8 percent have experienced intimate partner violence, and 20.3 percent have experienced violence perpetrated by another family member (INEC 2019). The economic costs of violence against women have been estimated at US$4.6 billion annually, equivalent to 4.3 percent of GDP (Vara-Horna 2020). Road traffic crashes impose a similarly large burden. According to the World Bank, they generated socioeconomic costs of approximately US$5.5 billion in 2022 (around 5 percent of GDP) and rank among the country's leading causes of death (Lopez Rodriguez et al. 2025). These figures illustrate the magnitude of the social costs potentially affected by temporary restrictions on alcohol availability.

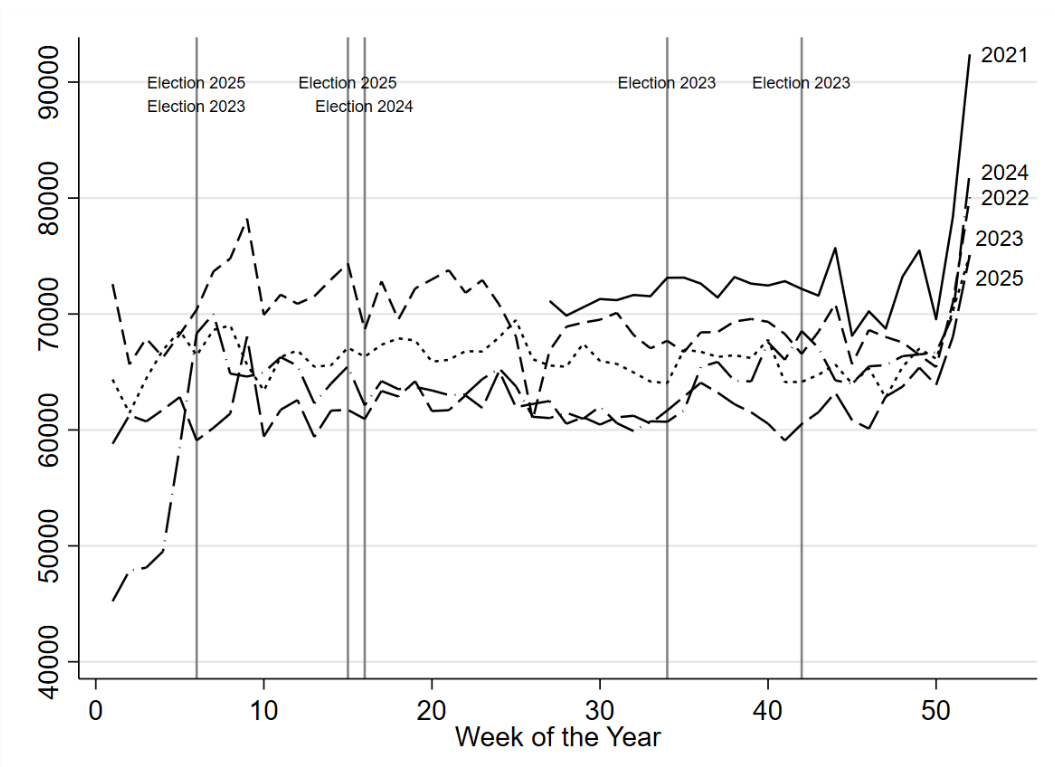
2.2 Data

Our analysis combines three nationwide administrative datasets obtained from Ecuadorian government institutions. The first dataset comes from ECU-911, Ecuador's integrated emergency response system, which receives emergency calls related to crimes, medical emergencies, traffic accidents, fires, and other incidents. We use the universe of emergency calls recorded between 2021 and 2025, including approximately 15.5 million observations. From these records, we identify 1.33 million alcohol-related incidents and 430,693 domestic and family violence incidents. These outcomes allow us to examine both the immediate effects of the alcohol ban on alcohol-related behavior and its consequences for domestic violence. The second dataset consists of recorded traffic accidents collected by the National Transit Agency (ANT). The database contains 172,656 accidents reported between 2018 and 2025 and includes detailed information on accident severity and the contributing causes, allowing us to identify crashes in which alcohol or drug impairment was recorded as the likely cause. The third dataset contains approximately 7.7 million hospital admissions recorded by public and private hospitals and compiled by Ecuador's National Institute of Statistics and Census (INEC) between 2018 and 2024. Using the primary diagnosis, we identify hospital admissions resulting from external causes, including traffic injuries, assaults, falls, and other traumatic injuries. These records provide a measure of severe health outcomes that complements the information obtained from emergency calls and the transit agency records.

All three datasets contain detailed geographic identifiers. For the empirical analysis, individual records are aggregated to the canton-day level, where cantons correspond to the second administrative division of Ecuador and are equivalent to municipalities. The final panel covers Ecuador's 221 cantons over multiple years and multiple electoral events.

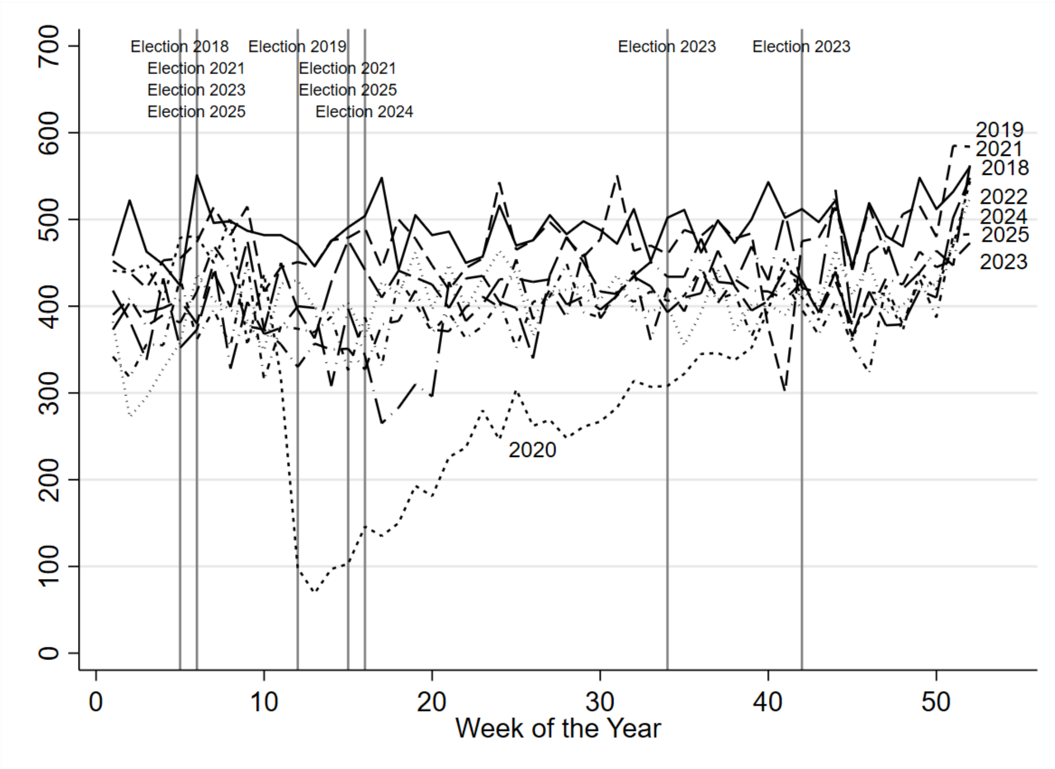
Figures 1–3 present descriptive evidence for the main outcome variables. Emergency calls, traffic accidents, and hospital admissions display stable weekly patterns across years. Appendix Figure A.1 further shows that alcohol-related incidents and domestic violence emergency calls follow similar daily dynamics, with both increasing during weekends, providing descriptive evidence consistent with the empirical strategy and the proposed mechanism.

Figure 1. Emergency Service Calls.



Note: Figures show the total number of emergency calls by week of the year, covering 2021 - 2025.

Figure 2. Recorded Traffic Accidents.

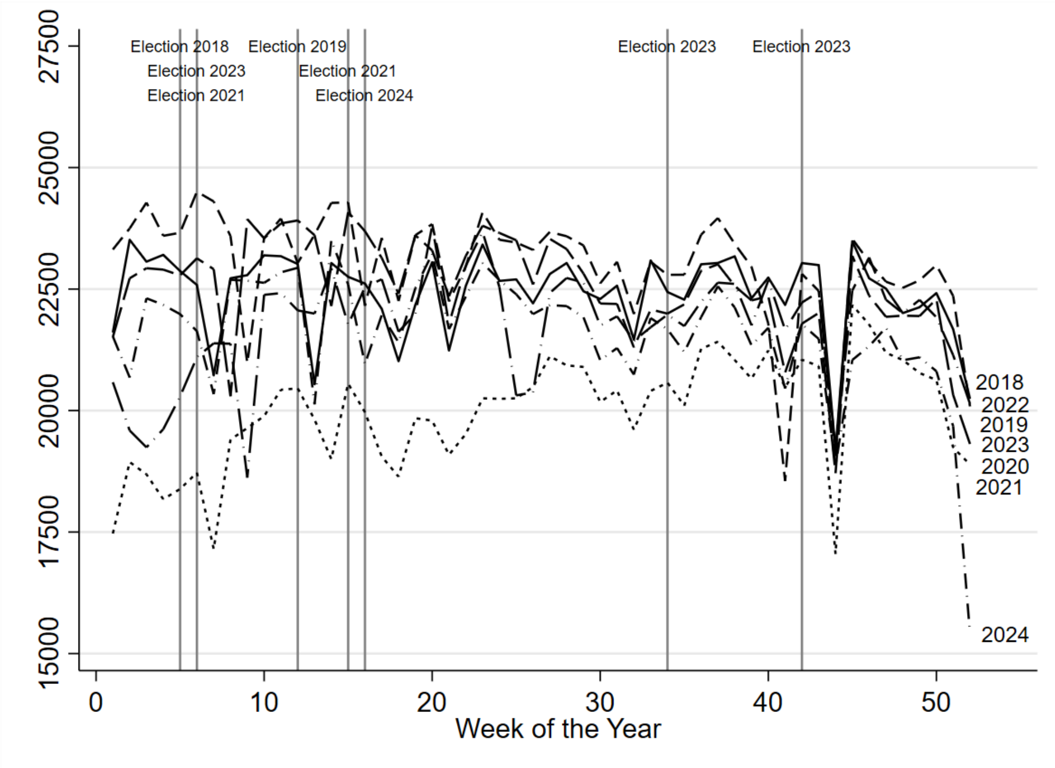


Note: Figures show the total number of recorded traffic accidents by week of the year, covering 2018 - 2025.

3 Empirical Strategy

Our empirical strategy exploits the timing of Ecuador's electoral alcohol bans. Figure 4 illustrates the difference-in-differences framework. The pre-treatment period corresponds to Monday through Thursday, whereas the post-treatment period covers Friday through Sunday, when the alcohol ban is in effect. Importantly, elections are held only on Sundays, while the ban starts on Friday, generating two days during which alcohol availability changes in the absence of electoral activities. Treated observations correspond to election weeks, while control observations are drawn from the same calendar week in all years in which no election takes place during that particular week. Identification therefore comes from comparing the within-week change in outcomes from Monday–Thursday to Friday–Sunday during election weeks with the corresponding within-week change in matched control weeks.

Figure 3. Hospital Admissions.

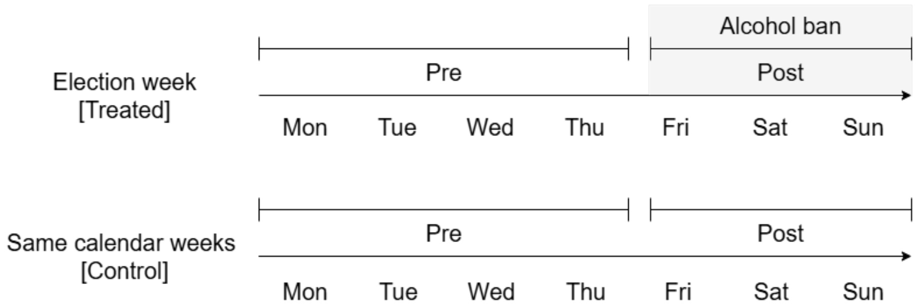


Note: Figures show the total number of hospital admissions by week of the year, covering 2018 - 2024.

Our approach is conceptually similar to recent studies exploiting temporal shocks and high-frequency administrative data to estimate causal effects by comparing affected periods with the same periods in previous years (Leslie & Wilson 2020, Abrams 2021). Relative to these studies, our setting provides ten independent shocks generated by electoral alcohol bans implemented between 2018 and 2025. To isolate the effects of the alcohol ban from unrelated weekly and seasonal changes, we control for systematic variation using day-of-week, week-of-year, and year fixed effects. Since elections are held exclusively on Sundays, any significant effects observed on Friday and Saturday are unlikely to be driven by election-related activities. In addition, we later present estimates excluding Sundays altogether, providing conservative estimates of the effects of alcohol restrictions.

We estimate the following difference-in-differences specification at the canton-day level:

Figure 4. Treatment and Control Weeks in the Difference-in-Differences Design



Note: The pre-treatment period corresponds to Monday–Thursday, while the post-treatment period covers Friday–Sunday, when the alcohol ban is in effect. Elections are held only on Sundays. For each election week, control observations consist of the same calendar week in all years in which the same calendar week is not affected by an election.

$$Y_{cwy,d} = \alpha + \beta \left(Treat_{wy} \times Post_d \right) + \lambda_{cwy} + \delta_d + \varepsilon_{cwy,d}, \tag{1}$$

where $Y_{cwy,d}$ represents the outcome in canton (c), week-of-the-year (w), year (y), and day-of-the-week (d). The variable $Treat_{wy}$ equals one for weeks with an electoral event (i.e., alcohol ban during the weekend) and zero for the control weeks. The variable $Post_d$ equals one for Friday through Sunday and zero for Monday through Thursday. The variable λ_{cwy} represents fixed effects at canton × week-of-the-year × year level, which controls for seasonal and annual variation across cantons. Day-of-the-week fixed effects, δ_d, account for systematic weekly patterns in the outcome variable. The coefficient of interest is β, which measures the differential change in outcomes during the alcohol-ban period in election weeks relative to matched non-election weeks. The main outcome variables we examine are alcohol-related and domestic violence emergency calls. Additionally, we also examine the effect on injury-related hospitalizations, and car accidents. The assumption for identification is parallel trends, meaning that outcome variables (i.e., domestic violence) would have had the same trend during weekends in the absence of the alcohol ban.

We also estimate event-study specifications to examine the dynamics of the effects over the election week:

$$Y_{cwy,d} = \sum_{k \neq -1} \beta_k \left(Treat_{wy} \times \mathbf{1} Day_d = k \right) + \lambda_{cwy} + \delta_d + \varepsilon_{cwy,d}, \tag{2}$$

where k indexes days relative to the start of the alcohol ban, with Thursday ($k = -1$) omitted as the reference category. The coefficients β_k trace the evolution of outcomes before and after the beginning of the ban. This specification allows us to assess the validity of the identifying assumption by examining pre-treatment differences during Monday–Thursday and to determine whether effects emerge precisely when the alcohol ban begins. In the robustness section, we assess the sensitivity of the results to alternative specifications, placebo exercises, and inference procedures, including wild cluster bootstrap methods. We also estimate specifications excluding Sundays to isolate the effects of alcohol restrictions from election-day activities.

4 Results

4.1 Effects on Alcohol-Related Incidents and Domestic Violence

We begin by examining whether the temporary alcohol ban reduced alcohol-related incidents and domestic violence, the two primary outcomes of the analysis. Table 2 reports the main difference-in-differences estimates. Column (1) shows that the alcohol ban reduced alcohol-related emergency calls by 1.20 incidents per canton-day (95% CI: $-2.03, -0.37$), corresponding to a 33 percent decline relative to the sample mean of 3.66 calls. Aggregated across Ecuador's 221 cantons, this estimate implies approximately 265 fewer alcohol-related emergency calls per day during alcohol-ban weekends. Column (2) shows that the reduction in alcohol-related incidents was accompanied by a substantial decline in domestic violence. The alcohol ban reduced domestic violence emergency calls by 0.33 incidents per canton-day (95% CI: $-0.44, -0.22$), equivalent to a 28.5 percent reduction relative to the sample mean. At the national level, this corresponds to approximately 73 fewer domestic violence emergency calls each day during the ban. These results indicate that temporary restrictions on alcohol availability substantially reduced both alcohol-related incidents and domestic violence.

Figure 5 examines the timing of these effects using the event-study specification. Panel (a)

Table 2. Effect of the alcohol ban on emergency service calls

	Alcohol Related	Domestic Violence
DiD	-1.198***	-0.330***
	[-2.025, -0.371]	[-0.440, -0.220]
Observations	34,034	34,034
Dep. var. mean	3.655	1.156

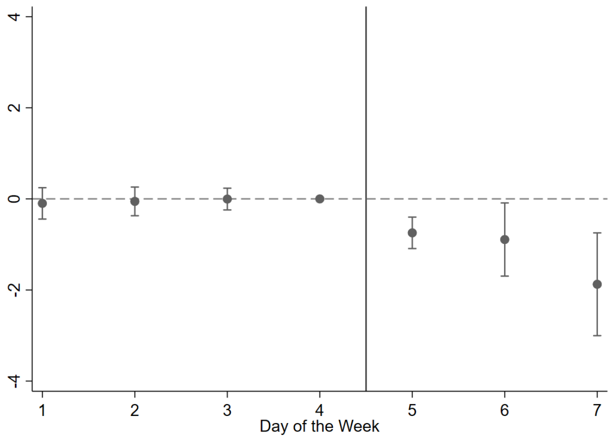
Notes: The table reports difference-in-differences estimates of the alcohol sales ban on emergency calls 2021 - 2025 (excluding 2020). p-values: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

shows that alcohol-related emergency calls remain statistically indistinguishable between treated and control weeks during the pre-treatment period (Monday–Thursday), providing support for the parallel trends assumption. Once the alcohol ban begins on Friday, alcohol-related emergency calls decline immediately and remain significantly below their baseline levels throughout the weekend. Panel (b) reports a similar pattern for domestic violence emergency calls. Pre-treatment coefficients are close to zero, while the effects are negative and significant precisely when the alcohol ban takes effect on Friday and persist through Sunday. The close correspondence between the dynamics of alcohol-related incidents and domestic violence is consistent with the proposed mechanism, where the temporary restriction on alcohol availability reduces alcohol-related behaviors, which in turn contributes to lower levels of domestic violence during election weekends. Given that elections take place only on Sunday whereas the alcohol ban begins on Friday, the significant reductions observed on Friday and Saturday are unlikely to be driven by electoral activities themselves.

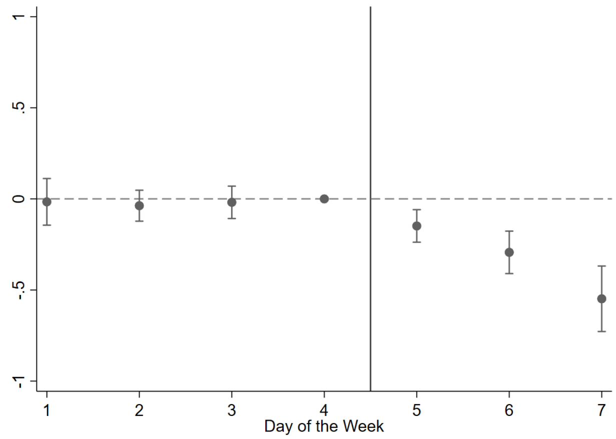
4.2 Heterogeneous Effects

We next examine whether the effects of the alcohol ban vary across locations with different levels of alcohol exposure, baseline prevalence of gender-based violence, and socioeconomic conditions. These analyses provide evidence on the settings in which temporary restrictions on alcohol availability are most effective and help identify potential sources of heterogeneity in the estimated average treatment effects. Throughout this section, heterogeneous effects are estimated using a

Figure 5. Event studies for the Emergency Service Calls.



(a) *Alcohol Related.*



(b) *Domestic Violence.*

Note: Figures show the event-study estimates of the alcohol sales ban on emergency calls 2021 - 2025 (excluding 2020), along with 95% confidence intervals.

triple-difference (DDD) specification.

4.2.1 Alcohol Expenditure

Our first heterogeneity analysis considers household alcohol expenditure, which serves as a proxy for baseline exposure to alcohol consumption. We classify cantons according to whether the average monthly household expenditure on alcohol in their corresponding province is above or below the national average, using information from the 2024–2025 National Survey of Household Income and Expenditure. Table 3 reports summary statistics for the two groups.

Table 3. Summary Stats by Alcohol Expenditure per Household at Province Level

	Alcohol Exp. per HH.	Alcohol Emergency	Domestic Violence
Below Average	2.303	4.136	1.110
Above Average	4.202	3.050	1.213

Notes: The table reports the means of variables by the average alcohol expenditure of households per month.

Table 4 presents the corresponding DDD estimates. The alcohol ban reduces alcohol-related emergency calls in both groups, although the estimated effects are somewhat larger in cantons belonging to provinces with below-average alcohol expenditure. Domestic violence also declines significantly in both groups. Estimated effects amount to 0.35 fewer domestic violence emergency calls per canton-day (95% CI: $-0.51, -0.20$) in below-average alcohol expenditure provinces and 0.30 fewer calls (95% CI: $-0.45, -0.15$) in above-average provinces, corresponding to reductions of approximately 32 and 25 percent relative to the respective group means. Overall, the results indicate that temporary alcohol bans reduce domestic violence regardless of baseline alcohol expenditure, although the estimated effects are somewhat larger in provinces with lower household alcohol expenditure.

Table 4. Emergency Service Calls by Alcohol Expenditure Level of Province

	Alcohol Emergency	Domestic Violence
Below Average	-1.639***	-0.353***
	[-2.440, -0.837]	[-0.509, -0.196]
Above Average	-0.645	-0.302***
	[-2.215, 0.925]	[-0.454, -0.149]
Observations	34,034	34,034
Dep. var. mean	3.655	1.156

Notes: The table reports triple difference-in-differences estimates of the alcohol sales ban on emergency calls 2021 - 2025. p-values: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

4.2.2 Baseline Prevalence of Gender-Based Violence

We next investigate whether the effectiveness of the alcohol ban depends on the underlying prevalence of violence against women. Using the 2019 National Survey on Family Relations and Gender Violence Against Women, we classify cantons according to whether the lifetime prevalence of violence against women in their corresponding province is above or below the national average. Summary statistics are reported in Table 5.

Table 6 reports the DDD estimates. Alcohol-related emergency calls decline in both groups, but the effects are substantially larger in provinces with higher baseline prevalence of gender-based violence. In these provinces, alcohol-related emergency calls decrease by 1.77 incidents per canton-day (95% CI: $-2.69, -0.85$), equivalent to a 43 percent reduction relative to the group mean, compared with a 21 percent reduction in provinces with lower prevalence. Domestic violence

Table 5. Summary Stats by Quantile of Gender Violence of Province

Gender violence level	Share of women experienced violence	Alcohol Emergency	Domestic Violence
Below Average	0.154	3.229	1.046
Above Average	0.278	4.125	1.277

Notes: The table reports the means of variables by share of the population below the poverty line.

exhibits a similar pattern. Emergency calls decline by 0.45 incidents per canton-day (95% CI: $-0.63, -0.26$), corresponding to a 35 percent reduction, in provinces with above-average prevalence of violence against women, compared with a 22 percent reduction in the remaining provinces. The similarity between the heterogeneous effects on alcohol-related incidents and domestic violence further supports the mechanism linking alcohol availability to domestic violence and suggests that temporary alcohol restrictions are particularly effective in areas where gender-based violence is more prevalent.

Table 6. Emergency Service Calls by Gender Violence Level of Province

	Alcohol Emergency	Domestic Violence
Below Average	-0.681	-0.226***
	[-2.019, 0.657]	[-0.355 -0.097]
Above Average	-1.769***	-0.445***
	[-2.687 -0.851]	[-0.628 -0.262]
Observations	34,034	34,034
Dep. var. mean	3.655	1.156

Notes: The table reports triple difference-in-differences estimates of the alcohol sales ban on emergency calls 2021 - 2025. p-values: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

4.2.3 Socioeconomic Conditions

Finally, we examine whether treatment effects differ across local socioeconomic conditions. We consider two complementary indicators measured at the canton level using the 2022 Population Census: multidimensional poverty, measured through the Unsatisfied Basic Needs Index, and educational attainment, measured as the share of the adult population with higher education.

Because poverty and educational attainment are only moderately correlated across cantons (Table A.1), both variables are examined separately.

Table 8 reports heterogeneous effects by poverty level. The reduction in alcohol-related emergency calls is concentrated in low-poverty cantons, where calls decrease by 3.60 incidents per canton-day (95% CI: $-6.00, -1.21$), equivalent to a 41 percent reduction relative to the group mean. The estimated effects are substantially smaller and statistically insignificant in medium- and high-poverty cantons. Domestic violence follows a similar pattern. Emergency calls decline by 0.82 incidents per canton-day (95% CI: $-1.13, -0.51$), corresponding to a 32 percent reduction, in low-poverty cantons, while the estimated declines in medium- and high-poverty cantons are considerably smaller. One possible explanation is that relatively more affluent cantons experience higher baseline levels of alcohol-related incidents and therefore respond more strongly to temporary restrictions on alcohol availability. However, differences in reporting behavior, enforcement, or drinking patterns may also contribute to the observed heterogeneity.

Table 7. Summary Stats by Quantile of Poverty Level of Canton

Poverty level	Share of HH. in poverty	Alcohol Emergency	Domestic Violence
Low	0.334	8.810	2.539
Med	0.518	1.261	0.518
High	0.750	0.855	0.401

Notes: The table reports the means of variables by share of the population below the poverty line.

Table 10 reports heterogeneous effects by educational attainment. The alcohol ban reduces alcohol-related emergency calls across all education groups, with statistically significant effects in medium- and high-education cantons. Relative to the corresponding group means, alcohol-related emergency calls decline by approximately 16 percent in low-education cantons, 45 percent in medium-education cantons, and 35 percent in high-education cantons. Domestic violence emergency calls also decline significantly across all groups, with reductions of approximately 21, 39, and 24 percent, respectively. Overall, these findings indicate that temporary alcohol bans reduce both alcohol-related incidents and domestic violence throughout the socioeconomic distribution,

Table 8. Emergency Service Calls by Poverty Level of Canton

	Alcohol Emergency	Domestic Violence
DDD Low	-3.603***	-0.816***
	[-5.995, -1.211]	[-1.127, -0.506]
DDD Med	-0.162	-0.085**
	[-0.437, 0.113]	[-0.160, -0.009]
DDD High	0.190	-0.086**
	[-0.040, 0.419]	[-0.154, -0.018]
Observations	34,034	34,034
Dep. var. mean	3.655	1.156

Notes: The table reports triple difference-in-differences estimates of the alcohol sales ban on emergency calls 2021 - 2025. p-values: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

although the largest relative effects are observed in medium-education and low-poverty cantons.

Table 9. Summary Stats by Quartile of Education Level of Canton

Education level	Share of Pop. with further Edu.	Alcohol Emergency	Domestic Violence
Low	0.568	3.318	1.371
Med	0.646	4.004	1.362
High	0.709	3.642	0.730

Notes: The table reports the means of variables by share of the population with further education.

Table 10. Emergency Service Calls by Education Level of Canton

	Alcohol Emergency	Domestic Violence
DDD Low	-0.529	-0.286***
	[-2.595, 1.536]	[-0.481, -0.092]
DDD Medium	-1.790***	-0.530***
	[-3.007, -0.573]	[-0.780, -0.281]
DDD High	-1.276***	-0.172***
	[-1.867, -0.684]	[-0.263, -0.081]
Observations	34,034	34,034
Dep. var. mean	3.655	1.156

Notes: The table reports triple difference-in-differences estimates of the alcohol sales ban on emergency calls 2021 - 2025. p-values: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

4.3 Broader Effects on Road Safety and Health

The previous results show that temporary alcohol bans substantially reduce alcohol-related incidents and domestic violence. We next examine whether these effects extend to other alcohol-related externalities by analyzing traffic accidents and external injury hospital admissions.

4.3.1 Traffic Accidents

Table 11 reports the effects of the alcohol ban on registered traffic accidents. Column (1) shows that total traffic accidents declined by 0.05 accidents per canton-day (95% CI: $-0.08, -0.03$), corresponding to an approximately 20 percent reduction relative to the sample mean. Aggregated across Ecuador's 221 cantons, this implies roughly 12 fewer traffic accidents each day during alcohol-ban weekends. The effects are considerably larger for accidents explicitly attributed by the police to alcohol or drug impairment. Column (2) shows a reduction of 0.015 alcohol-related accidents per canton-day (95% CI: $-0.02, -0.01$), equivalent to approximately three fewer alcohol-related accidents nationwide each day and an 83 percent decline relative to the sample mean. Because alcohol involvement may not always be identified or recorded by police officers, this estimate should be interpreted as a lower bound of the true effect of the alcohol ban on alcohol-related crashes.

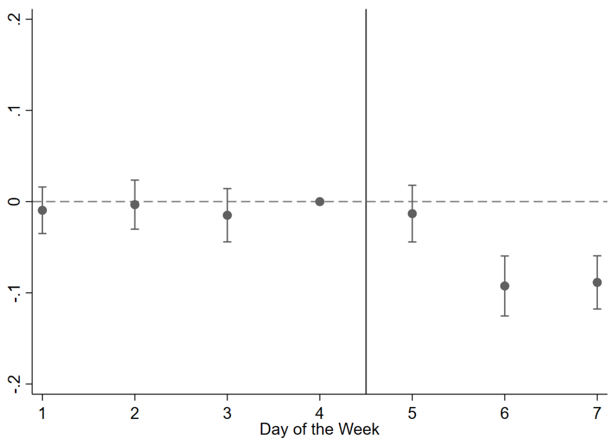
Table 11. Recorded Traffic Accidents.

	All Accidents	Alcohol Related
DiD	-0.052***	-0.015***
	[-0.078, -0.027]	[-0.021, -0.009]
Observations	76,489	76,489
Dep. var. mean	0.256	0.018

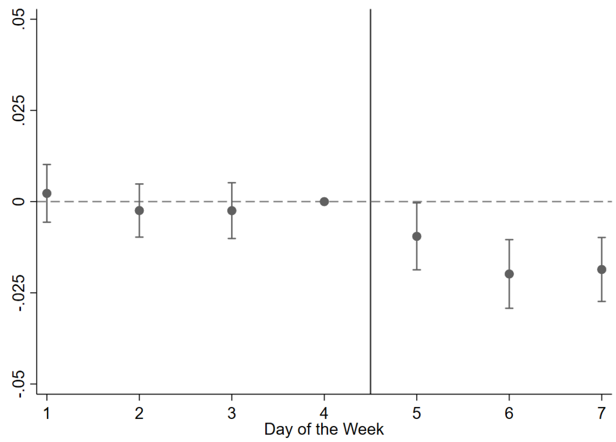
Notes: The table reports difference-in-differences estimates of the alcohol sales ban on recorded traffic accidents 2018 - 2025 (excluding 2020). p-values: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Figure 6 presents the corresponding event-study estimates. Total traffic accidents begin to decline on Saturday and remain below their usual levels on Sunday. In contrast, alcohol-related traffic accidents decrease immediately when the alcohol ban takes effect on Friday, with negative and statistically significant effects persisting throughout the weekend. These dynamics closely mirror those observed for alcohol-related emergency calls, providing additional evidence that the policy reduced alcohol-related behaviors.

Figure 6. Event studies for the Recorded Traffic Accidents.



(a) *All Accidents.*



(b) *Alcohol Related..*

Note: Figures show the event-study estimates of the alcohol sales ban on recorded traffic accidents 2018 - 2025 (excluding 2020), along with 95% confidence intervals.

Table 12 further examines heterogeneity by accident severity. The alcohol ban significantly reduced all categories of traffic accidents. Fatal crashes declined by 0.010 accidents per canton-day (95% CI: $-0.015, -0.005$), corresponding to approximately two fewer fatal crashes nationwide each day. Injury crashes decreased by 0.025 accidents per canton-day (95% CI: $-0.040, -0.010$), equivalent to roughly six fewer injury crashes per day, while property-damage-only crashes declined by 0.018 accidents per canton-day (95% CI: $-0.034, -0.003$), or approximately four fewer crashes nationwide each day. These results indicate that the benefits of temporary alcohol bans extend across the full distribution of crash severity.

4.3.2 External Injury Hospital Admissions

We next examine whether the alcohol ban affected hospital admissions related to external injuries. Unlike emergency calls or police records, hospital admissions capture more severe health conse-

Table 12. Traffic Accidents by Outcome.

	Fatality One or more	Injury Only	Property Damage Only
DiD	-0.010***	-0.025***	-0.018**
	[-0.015, -0.005]	[-0.042, -0.008]	[-0.034, -0.003]
Observations	75,117	76,146	72,373
Dep. var. mean	0.022	0.142	0.098

Notes: The table reports difference-in-differences estimates of the alcohol sales ban on recorded traffic accidents 2018 - 2025 (excluding 2020). p-values: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

quences associated with alcohol-related risk-taking, including traffic injuries, assaults, falls, and other accidental injuries.

Table 13 reports the main estimates. The alcohol ban significantly reduced admissions for external injuries while having no detectable effect on either total hospital admissions or admissions unrelated to injuries. Column (3) shows that external injury admissions declined by 0.128 admissions per canton-day (95% CI: $-0.256, -0.001$), corresponding to a 5.3 percent reduction relative to the sample mean. The absence of effects on overall hospital admissions suggests that the policy specifically reduced injuries plausibly associated with alcohol consumption rather than hospital utilization more generally.

Table 13. Hospital Admissions.

	Overall Admissions	External Injury Related
DiD	-0.070	-0.128**
	[-1.993, 1.852]	[-0.256, -0.001]
Observations	37,338	37,338
Dep. var. mean	23.988	2.379

Notes: The table reports difference-in-differences estimates of the alcohol sales ban on hospital admission 2018 - 2024 (excluding 2020). p-values: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 14 explores heterogeneity by gender and age. The reduction in injury-related admissions is concentrated among men and among individuals aged 15–30 and 30–60 years. Figure 7 presents the corresponding event-study estimates. Panel (a) shows that injury admissions among men decline sharply beginning on Saturday, whereas the estimates for women remain small and statistically insignificant throughout the weekend. This pattern is consistent with national survey evidence indicating substantially higher alcohol consumption among men than women in Ecuador. According to the National Health and Nutrition Survey, 51.8 percent of men consumed alcohol during the previous month compared with 27.3 percent of women, while harmful alcohol consumption among drinkers was also considerably more prevalent among men (36.0 percent) than women (11.9 percent) (Ministerio de Salud Publica 2018). Panel (b) shows that the reduction in injury-related admissions is concentrated among individuals aged 15–30 and 30–60 years. For both age groups, admissions decline beginning on Saturday and remain below their typical weekend levels. In contrast, the estimated effects for individuals aged over 60 are small and statistically insignificant throughout the weekend. Overall, these findings suggest that temporary alcohol bans primarily reduce injuries among demographic groups with higher baseline alcohol consumption and greater exposure to alcohol-related risk-taking.

Table 14. External Injury Related Hospital Admissions.

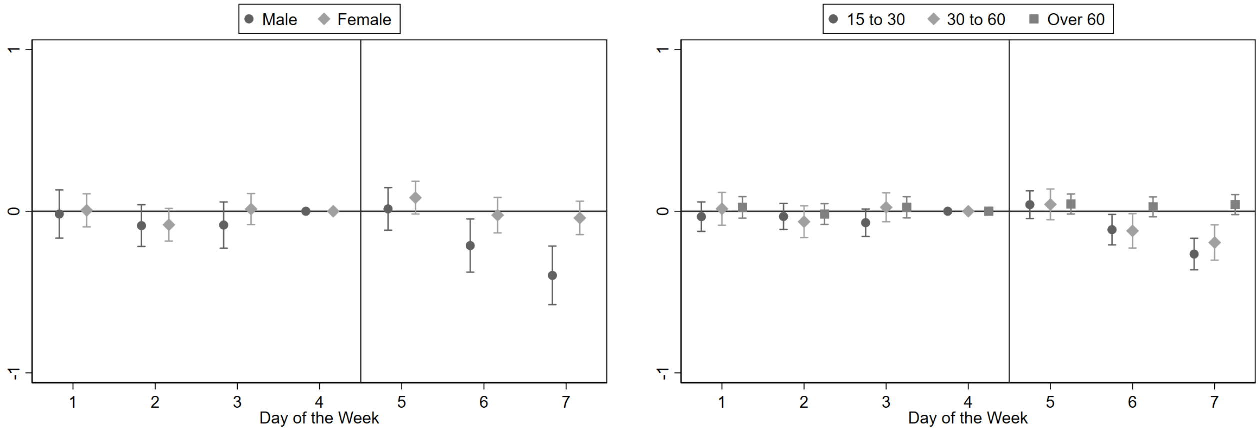
	Female	Male	15 to 30	30 to 60	Over 60
DiD	0.022	-0.150***	-0.078***	-0.085 ***	0.030*
	[-0.041, 0.085]	[-0.242, -0.058]	[-0.134 -0.023]	[-0.149, -0.021]	[-0.005, 0.065]
Observations	37,338	37,338	37,338	37,338	37,338

Notes: The table reports difference-in-differences estimates of the alcohol sales ban on hospital admission 2018 - 2024 (excluding 2020). p-values: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

4.4 Economic Benefits

The estimated reductions in domestic violence and traffic accidents also have important economic implications. In this section, we translate our estimated treatment effects into monetary values using existing estimates of the costs of domestic violence and road traffic crashes in Ecuador. Because these calculations combine our causal estimates with external cost estimates and simplifying assumptions, they should be interpreted as illustrative measures of the short-run economic benefits of temporary alcohol bans rather than comprehensive welfare analyses.

Figure 7. Event Studies for the External Injury-related Hospital Admissions.



(a) *Gender*

(b) *Age Bracket*

Note: Figures show the event-study estimates of the alcohol sales ban on external injury related hospital admission 2018 - 2024 (excluding 2020), along with 95% confidence intervals.

4.4.1 Domestic Violence

Table 15 presents a back-of-the-envelope estimate of the direct short-term economic savings associated with the reduction in domestic violence during alcohol-ban weekends. We use the cost estimates reported by Vara-Horna (2020), which provides the most comprehensive assessment of the economic costs of violence against women in Ecuador. From the full set of costs, we retain only those categories that are expected to respond in the short run, namely women's lost income and time, out-of-pocket expenditures on health care, justice and protection, children's health-care expenditures, and direct public health-care costs.

To estimate the implied savings during an alcohol-ban weekend, annual costs are converted to a three-day equivalent and multiplied by our estimated 28.5 percent reduction in domestic violence reported in Table 2. The calculation suggests direct short-term savings of approximately US$3.22 million per alcohol-ban weekend. The largest component corresponds to avoided losses in women's income and time, accounting for approximately US$2.65 million (82 percent of total savings), followed by reductions in children's health-care expenditures. These estimates should be interpreted as illustrative lower-bound estimates. They assume that the selected cost categories vary proportionally with short-run changes in domestic violence and exclude numerous longerterm consequences, including psychological harm, productivity losses beyond the election weekend, criminal justice expenditures, and the long-run effects of violence on women and children.

Table 15. Estimated Domestic Violence Direct Short-Term Cost Savings During an Alcohol-Ban Weekend

Level	Direct short-term costs	Cost USD per weekend	Savings USD per weekend
Individual	Women's lost income and time	9,310,086	2,653,374
Individual	Health care, justice, and protection	555,115	158,208
Household	Children's health care costs	1,281,478	365,221
Public sector	Direct health care costs	162,179	46,221
Total cost		11,308,859	3,223,025

Notes: Calculations use selected direct short-term cost categories reported by (Vara-Horna 2020). Annual costs are converted to a three-day equivalent by multiplying by 3/365 and then multiplied by the estimated 28.5 percent reduction in domestic violence during alcohol-ban weekends. Estimates exclude long-term, indirect, and less immediately adjustable costs and should therefore be interpreted as illustrative lower-bound estimates of direct short-term savings.

4.4.2 Traffic Accidents

Table 16 presents an illustrative estimate of the economic savings associated with the reduction in traffic accidents during alcohol-ban weekends. The calculation combines our estimated reduction in traffic accidents with the unit economic costs of road crashes reported by the recent World Bank assessment of the socioeconomic costs of road crashes in Ecuador (Lopez Rodriguez et al. 2025). The heterogeneous estimates presented in Table 12 imply that the alcohol ban prevents approximately 7 fatal crashes, 17 injury crashes, and 12 property-damage-only (PDO) crashes during a typical three-day election weekend. Applying the corresponding unit costs reported by the World Bank yields estimated savings ranging between US\$3.25 million and US\$4.01 million per alcohol-ban weekend. Most of these savings arise from avoided fatal crashes, which account for approximately US\$3.13 million. Injury crashes contribute between US\$0.12 million and US\$0.88 million, depending on whether injuries are classified as slight or serious, while avoided property-damage-only crashes account for an additional US\$9,344.

The two exercises suggest that temporary alcohol bans generate substantial short-term economic benefits during a single election weekend. These estimates should nevertheless be interpreted as conservative lower-bound benchmarks. First, the domestic violence calculations include

Table 16. Estimated Road Crash Cost Savings During an Alcohol-Ban Weekend

By severity	Cost USD per crash	Accidents avoided per weekend	Savings USD per weekend
Fatality	471,528	7	3,126,231
Serious or slight injury	$52,848 - 6,965$	17	$875,956 - 115,445$
Property damage only	783	12	9,344
Total	–	35	4,011,531 3,251,020

Notes: Savings are calculated by multiplying the estimated reduction in traffic accidents reported in Table 11 by the economic cost per crash reported by Lopez Rodriguez et al. (2025). The number of avoided crashes by severity assumes the same distribution as observed in the administrative accident records 2024-2026. The range for injury crashes reflects the World Bank's alternative estimates for slight and serious injuries.

only selected direct costs that are expected to adjust in the short run. Second, the traffic accident calculations do not include the economic benefits associated with the reduction in external injury hospital admissions documented in Table 13. We do not monetize these additional health benefits because many hospitalization costs associated with road traffic injuries are already incorporated into the World Bank crash cost estimates, raising the possibility of double counting. At the same time, external injury admissions also include injuries unrelated to road traffic crashes, such as falls, assaults, and other alcohol-related injuries. Consequently, the overall economic benefits of temporary alcohol bans are likely to exceed the estimates reported here.

5 Robustness Checks

5.1 Excluding Sundays

Our main specification includes the entire election weekend, during which the alcohol ban is in force from Friday through Sunday. Although the identifying assumption relies on the ban rather than the election itself, election-day activities on Sunday could potentially influence mobility, reporting behavior, or other outcomes independently of alcohol availability. To address this concern, we re-estimate our main specifications excluding Sundays and therefore identify the effects of the alcohol ban using only Friday and Saturday, when the restriction is in place but no electoral activities occur. Table 17 reports these estimates for emergency calls. The alcohol ban continues to generate statistically significant reductions in both alcohol-related incidents and domestic violence. The estimated effects remain economically large, with alcohol-related emergency calls declining by approximately 27 percent and domestic violence calls by 20.6 percent. Although these magnitudes are smaller than the baseline estimates, this difference is expected because Sundays exhibit the highest baseline incidence of both alcohol-related and domestic violence emergency calls (Figure A.1).

Table 17. Effect of the alcohol ban on emergency service calls - Excluding Sundays

	Alcohol Related	Domestic Violence
DiD	-0.825**	-0.215***
	[-1.493, -0.157]	[-0.302, -0.128]
Observations	29,172	29,172
Dep. var. mean	3.049	1.044

Notes: The table reports difference-in-differences estimates of the alcohol sales ban on emergency calls 2021 - 2025. p-values: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 18 shows that the reductions in traffic accidents are also robust to excluding Sundays. Estimated effects remain similar in magnitude to the baseline specification and are not statistically different from those obtained using the full weekend. In contrast, the estimated reduction in external injury hospital admissions remains negative but is no longer statistically significant. This result is consistent with the baseline weekly pattern of hospital admissions (Figure A.2), which typically decline on Fridays and Saturdays while peaking on Mondays, suggesting that some hospital admissions may occur with a delay following injuries sustained during the weekend. In summary, these results indicate that our main findings are not likely to be driven by election-day activities and instead reflect changes associated with the temporary restriction on alcohol availability.

5.2 Temporal Placebo Tests

To assess whether our estimates capture other within-week patterns unrelated to the alcohol ban, we conduct a placebo test by assigning a fictitious treatment to Wednesday and Thursday of election weeks, when no alcohol restriction is in place. Under the identifying assumptions of our

Table 18. Recorded Traffic Accidents - Excluding Sundays.

	All Accidents	Alcohol Related
DiD	-0.044***	-0.014***
	[-0.072, -0.016]	[-0.020, -0.008]
Observations	65,562	65,562
Dep. var. mean	0.242	0.016

Notes: The table reports difference-in-differences estimates of the alcohol sales ban on recorded traffic accidents 2018 - 2025 (excluding 2020). p-values: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 19. Hospital Admissions - Excluding Sundays.

	Overall Admissions	External Injury Related
DiD	0.090	-0.005
	[-1.499, 1.679]	[-0.124, 0.114]
Observations	32,004	32,004
Dep. var. mean	25.119	2.404

Notes: The table reports difference-in-differences estimates of the alcohol sales ban on hospital admission 2018 - 2024 (excluding 2020). p-values: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

design, no treatment effects should be observed during these days. Table 20 reports the placebo difference-in-differences estimates for emergency calls, while Figure 8 presents the corresponding event-study estimates. As expected, we find no statistically significant effects on either alcohol-related emergency calls or domestic violence. The event-study coefficients also remain close to zero throughout the placebo treatment period, providing no evidence of anticipatory behavioral responses prior to the implementation of the alcohol ban.

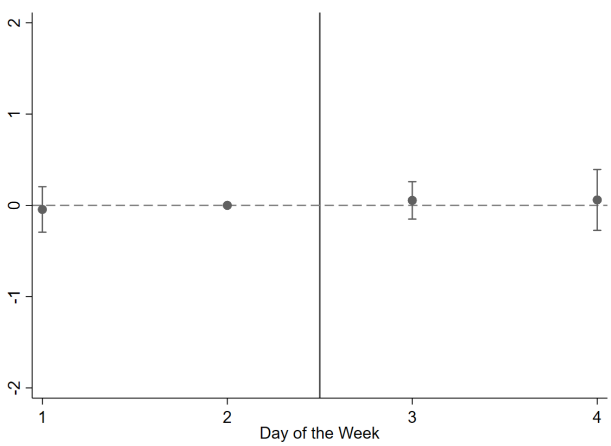
Tables 21 and 22 report analogous placebo estimates for traffic accidents and hospital admissions. Consistent with the emergency call results, we find no statistical placebo effects for either outcome. These findings support the identifying assumption that the estimated treatment effects are driven by the temporary alcohol ban rather than by unrelated within-week patterns or pre-existing trends.

Table 20. Effect of the alcohol ban on emergency service calls - Placebo One

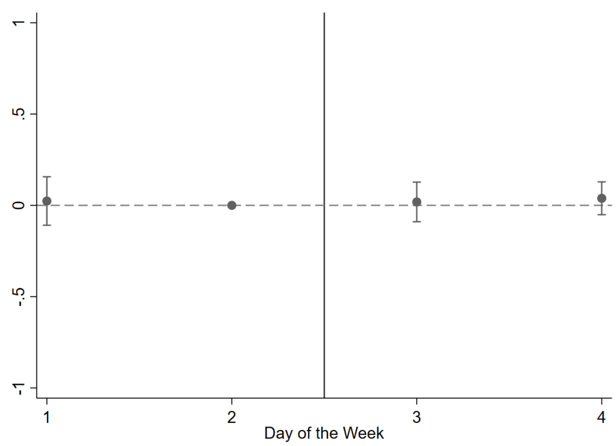
	Alcohol Related	Domestic Violence
DiD	0.079	0.017
	[-0.162, 0.321]	[-0.074, 0.107]
Observations	19,448	19,448
Dep. var. mean	2.148	0.967

Notes: The table reports difference-in-differences estimates of the alcohol sales ban on emergency calls 2021 - 2025. p-values: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Figure 8. Event studies for the Emergency Service Calls - Placebo One.



(a) *Alcohol Related.*



(b) *Domestic Violence.*

Note: Figures show the event-study estimates of the alcohol sales ban on emergency calls 2021 - 2025 (excluding 2020), along with 95% confidence intervals.

Table 21. Recorded Traffic Accidents - Placebo One.

	All Accidents	Alcohol Related
DiD	0.007	-0.001
	[-0.015, 0.029]	[-0.006, 0.004]
Observations	43,708	43,708
Dep. var. mean	0.216	0.011

Notes: The table reports difference-in-differences estimates of the alcohol sales ban on recorded traffic accidents 2018 - 2025 (excluding 2020). p-values: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 22. Hospital Admissions - Placebo One.

	Overall Admissions	External Injury Related
DiD	0.364	0.056
	[-0.183, 0.912]	[-0.069, 0.182]
Observations	21,336	21,336
Dep. var. mean	27.350	2.507

Notes: The table reports difference-in-differences estimates of the alcohol sales ban on hospital admission 2018 - 2024 (excluding 2020). p-values: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

6 Conclusion

Our findings show that temporary alcohol bans substantially reduce alcohol-related emergency calls, domestic violence, traffic accidents, and external injury hospital admissions. The immediate decline in alcohol-related incidents following the introduction of the ban, together with the parallel reductions in violence, injuries, and road traffic crashes, is consistent with reduced alcohol availability as the mechanism underlying these effects. We also find that the reduction in domestic violence is larger in locations with higher baseline levels of violence against women, while the decline in traffic accidents include both fatal and non-fatal crashes. Illustrative calculations indicate that a typical alcohol-ban weekend generates substantial short-term economic savings through reductions in domestic violence, primarily by avoiding losses in women's income and time; and in road traffic crashes, where avoided fatalities account for the largest share of the estimated benefits. Although these estimates rely on simplifying assumptions and should not be interpreted as comprehensive welfare measures, they illustrate that even temporary restrictions on alcohol availability can generate sizeable benefits for households, the public sector, and society more broadly.

Several limitations should be acknowledged. Our analysis identifies the short-run effects of a temporary and strongly enforced nationwide restriction and therefore does not imply that permanent or less strictly enforced alcohol policies would produce similar effects. Nevertheless, election-weekend alcohol bans belong to a broader class of time-based alcohol regulations, including late-night sales restrictions, event-specific prohibitions, and emergency public health measures, which are widely implemented across countries. Therefore, our findings have implications beyond the specific institutional setting examined here.

Overall, our findings provide causal evidence to inform the World Health Organization's SAFER initiative, which identifies restrictions on alcohol availability as the most cost-effective strategy for reducing alcohol-related harm (WHO 2025*a*). We also contribute to broader international efforts to inform evidence-based policies aimed at preventing violence against women and children (WHO 2026). More broadly, our results generate new evidence on the social externalities of alcohol consumption. While governments often justify alcohol-control policies by their potential to reduce harmful drinking, our findings show that their benefits extend well beyond alcohol consumption itself. Temporary restrictions on alcohol availability can generate important public health and social benefits by reducing domestic violence, traffic accidents, and injury-related morbidity, such externalities should be incorporated in evaluation analysis to better inform the design and implementation of alcohol regulation policies.

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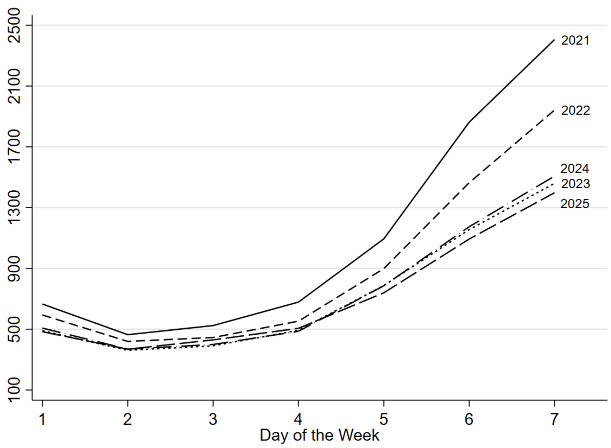
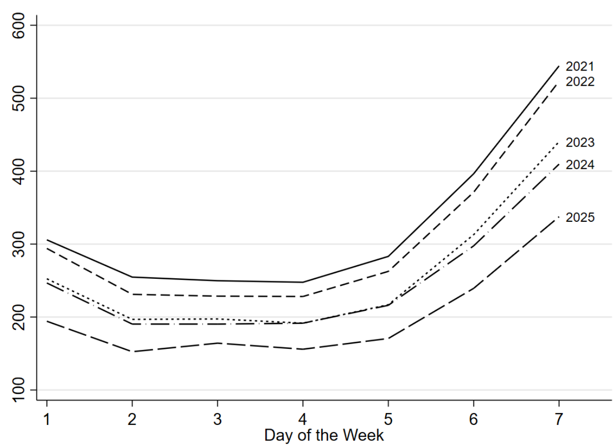
A Appendix

Figure A.1. Average Number of Emergency Service Calls.



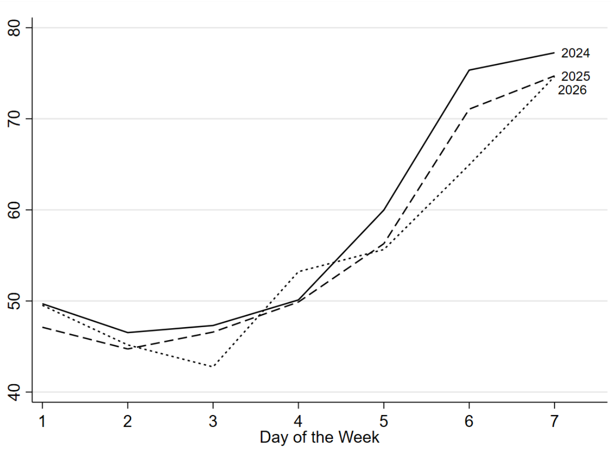
(a) *Alcohol Related.*



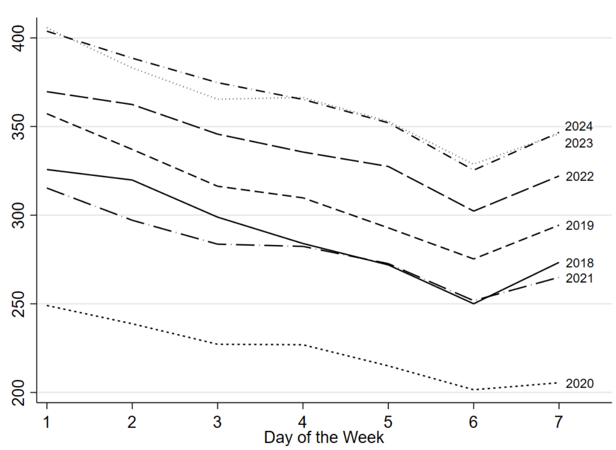
(b) *Domestic Violence.*

Note: Figures show the average number of emergency calls by week of the day, covering 2021 - 2025.

Figure A.2. Average Number of Recorded Traffic Accidents and Injury-Related Hospital Admissions.



(a) *Traffic Accidents.*



(b) *Injury Related Hospitalisations.*

Note: Figures show the average number of injury-related hospital admissions (2018 - 2024) and recorded traffic accidents (2018 - 2025) by week of the day.

Table A.1. Cantons by Poverty and Education Levels

	Education Level			
Poverty Level	Low	Med	High	Total
Low	19	29	26	74
Med	17	23	34	74
High	38	22	13	73
Total	74	74	73	221

Notes: The table reports the number of cantons by level of poverty and Education.